Group B: statistics and paper list

GPU sharing for multi-tenant inference | 16 October 2026

Read this first. The research is not fully screened yet. The current 15 QA/extraction forms are not a final list of 15 included studies.

Starting records by source

Source	Occurrences	Linked repeats	Canonical
ACM	1582	30	1552
IEEE	102	9	93
Springer	587	3	584
ScienceDirect	1898	67	1831
Supplementary	20	5	15
TOTAL	4189	114	4075

Narrowing and assessment coverage

Stage	Count
Imported occurrences	4189
Applied duplicate links	114
Canonical records	4075
Initial automated recommendations	4075
Preliminary exclusion proposals	1441
Retain or uncertain after initial triage	2634
Source-assessed reports	17
Provisionally eligible reports	13
Companion candidate reports	1
Publication-version-uncertain reports	1
Assessed full-text EC2 exclusions	2
QA and extraction forms	15
Q1-Q10 item rows	150
Reported outcome rows	45
Separate calculated outcomes	2
Unassessed canonical retain/uncertain records	2617
Final included reports/studies	Pending

Preliminary exclusion proposals: EC1=851; EC2=453; EC3=6; EC5=131. These 1,441 proposals are not final screened exclusions. The two source-assessed EC2 exclusions belong to the assessed report slice.

The 15 reports currently carrying QA/extraction

Six conference/journal DOIs are identified below. For the other nine reports, use the official/accessed PDF links in CURRENT_ASSESSED_PAPERS_AND_DOIS.csv. An unverified DOI is not invented.

Report / paper	DOI or source identifier	Current status
GB-P001 VastPipe (2025)	10.1007/s11390-024-3773-5	provisional
GB-P002 iGniter (2023)	10.1109/TPDS.2022.3232715	provisional
GB-P003 gpulets (2022)	Official PDF; DOI not verified	provisional
GB-P004 PipeSwitch (2020)	Official PDF; DOI not verified	provisional
GB-P005 REEF (2022)	Official PDF; DOI not verified	companion candidate
GB-P006 Paella (2023)	10.1145/3600006.3613163	provisional
GB-P007 Orion (2024)	10.1145/3627703.3629578	provisional
GB-P008 GSLICE (2020)	10.1145/3419111.3421284	provisional
GB-P009 Clockwork (2020)	Official PDF; DOI not verified	provisional
GB-P010 SIRIUS (2025)	Official PDF; DOI not verified	provisional
GB-P011 ParvaGPU (2024)	10.1109/SC41406.2024.00048	publication version uncertain
GB-P012 Torpor (2025)	Official PDF; DOI not verified	provisional
GB-P013 StreamBox (2024)	Official PDF; DOI not verified	provisional
GB-P014 KRYPTON (2025)	Official PDF; DOI not verified	provisional
GB-P015 XSched (2025)	Official PDF; DOI not verified	provisional

Source-assessed exclusions

Report	Decision / reason
GB-P016 TGS: Transparent GPU Sharing in Container Clouds for Deep Learning Workloads	EC2: Training-focused GPU container sharing; evaluation concerns production/opportunistic training jobs. Discussion explicitly places inference optimization outside this paper; no separable inference-sharing result identified.
GB-P017 HiveD: Sharing a GPU Cluster for Deep Learning with Guarantees	EC2: GPU-cluster resource reservation and affinity for deep-learning training; evaluated queueing/completion-time behavior is not concurrent local inference serving. EC2 decisive; aggregate cluster scope is a secondary limitation.

REEF: conference/journal study relationship is unresolved. ParvaGPU: the conference DOI is identified, but extracted evidence comes from the accessed arXiv version; final-publication equivalence is unresolved. Other assessments remain provisional.

What is complete, and what is pending?

Cycle	Current state	Remaining
1 Recovery and requirements audit	Closed audit scope	Research gaps tracked separately
2 Queries/imports/four-database audit	Closed audit scope with documented gaps	ACM run mapping; historical hit/export discrepancies; search completeness
3 Screening/full-text selection	Partial	2617 canonical candidates lack source assessment; final selection pending
4 Quality assessment	Partial overall; current15forms covered	23section/figure anchors;3provisional N/A calibrations; future eligible candidates
5 Data extraction	Partial overall; current15forms covered	Unchecked fields and new eligible candidates; version reconciliation
6 PRISMA and references	Interim outputs	Final selection and report/study linkage needed for final flow

Through instructor Heading 14

Heading	State	Specific remaining work
8 Search strategy	Partial	Query/export audit added; ACM 79-file mapping and MPS 517 displayed versus 563 parsed unresolved; planned serving synonyms and older MPS lanes require verification
9 Literature period	Partial	Per-run historical dates separated; publication_date_audit.csv flags missing/older/2026 dates for checks
10 Inclusion/exclusion	Partial	4075 initial triage recommendations;1441 exclude proposals;2634 retain/uncertain;17 source assessments and102 IC rows;complete abstract/source selection pending
11 PRISMA 2020	Partial	4189 occurrences minus114 links equals4075 canonical;2 assessed EC2 exclusions;final selection/PRISMA pending; current interim diagram embedded in current report
12 Search/screening log	Partial	4075 record-keyed screening rows and4189 occurrence-ledger rows; source decisions/IDs preserved; no human consensus or final included-study count claimed
13 Quality assessment	Partial	15 assessed forms/150 first QA items; source/calibration/selection gaps retained
14 Data extraction	Partial	Expand beyond current assessed set; resolve source versions and unchecked fields

Recorded dates and timestamps

All 4,075 screening rows have a recorded date after matching existing vote events. Exact event timestamps remain unrecorded for 798 rows. These stay “Not recorded”. A metadata-check timestamp is not substituted for an actual screening event. No searches, reviewer actions or times were fabricated.